

TS-ST-Overview-001

PolyFlop8

Interconnection and Verification Strategy for Hierarchical Testing

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REVISION HISTORY

Rev	Changes	Date
1.1	- Added <code>io_addr_in</code> (8-bit) to the Signal Dictionary in Appendix A. - Updated interface definitions to include the 8-bit connection between the Decoder and Register File. - Added change note for <code>io_addr_in</code> timing in the System-Level Verification Matrix.	06.01.2026
1.0	Initial architecture overview	06.01.2026

1 EXECUTIVE SUMMARY

1.1 Document Purpose

This document provides the high-level test architecture overview for the PolyFlop8 microcontroller verification. It defines the hierarchical test strategy, block interconnections at each verification stage, and establishes the framework for subsequent detailed test plans.

1.2 Scope

- Hierarchical test methodology from unit to system level
- Block interconnection definitions for each test stage
- Interface verification strategy
- Test environment architecture
- Cross-references to detailed test plans

1.3 Document Relationships

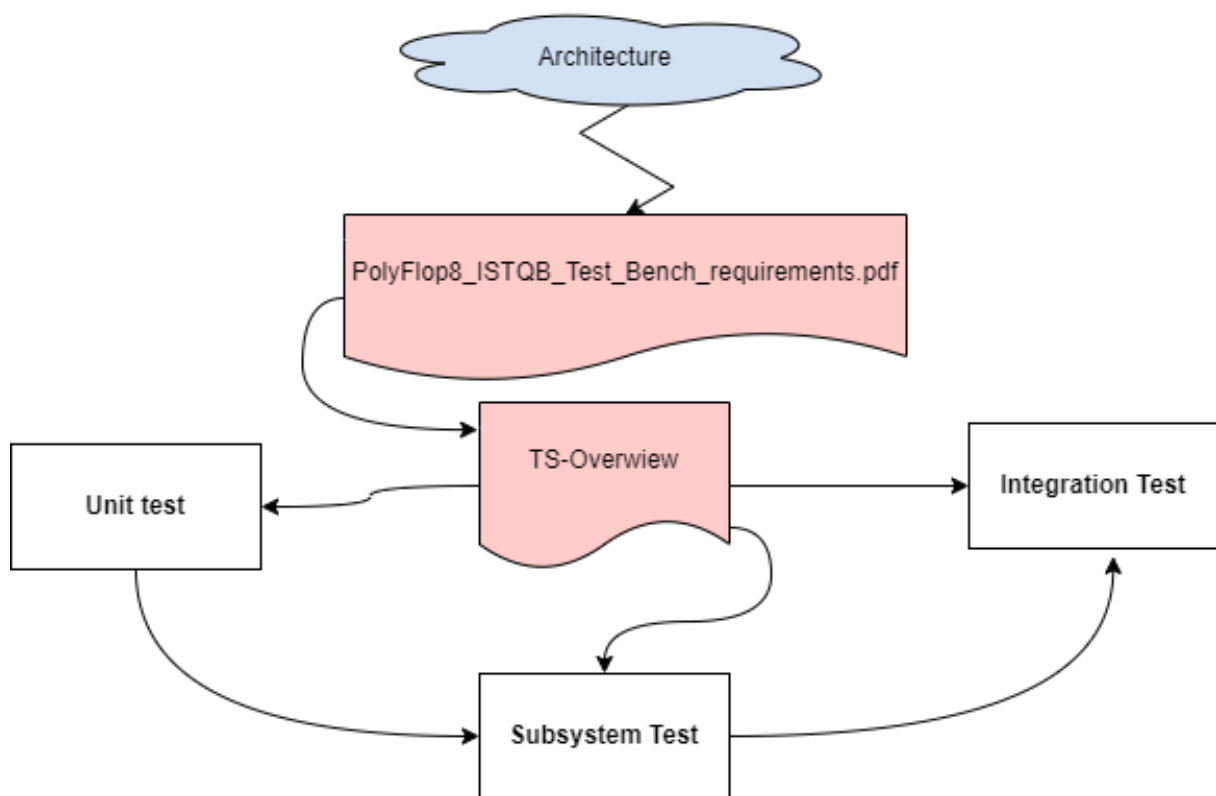


Figure 1: Document Relationships

2 TEST STRATEGY OVERVIEW

2.1 Hierarchical Verification Approach

The PolyFlop8 verification follows a bottom-up hierarchical approach with four distinct test stages:

Stage	Test Plan	Primary Focus	Coverage Target
Stage 1: Unit Level	Individual Block Tests	Standalone block functionality	95% block coverage
Stage 2: Data Path Integration	TS-ST-DP-001	ALU + RegFile + SREG interaction	90% path coverage
Stage 3: Fetch Unit Integration	TS-ST-FETCH-001	PC + ROM + IR + Decoder interaction	90% sequence coverage
Stage 4: Top-Level System	TS-ST-TOP-001	Full CPU with Control Unit	85% system coverage

2.2 Verification Methodology

- **Bottom-Up Integration:** Verified blocks are combined into subsystems
- **Interface-First Testing:** All block interfaces verified before functionality
- **Gradual Complexity:** Test complexity increases with integration level
- **Coverage-Driven:** Metrics guide test development and completion
- **Regression-Aware:** Tests organized for efficient regression suites

3 BLOCK INTERCONNECTIONS BY STAGE

3.1 Stage 1: Unit Level Testing

Unit testing serves as the foundation of the verification hierarchy. In this stage, individual hardware blocks (such as the ALU, Register File, and RAM) are verified in isolation to ensure they meet their specific functional requirements (e.g., truth tables, read/write timing).

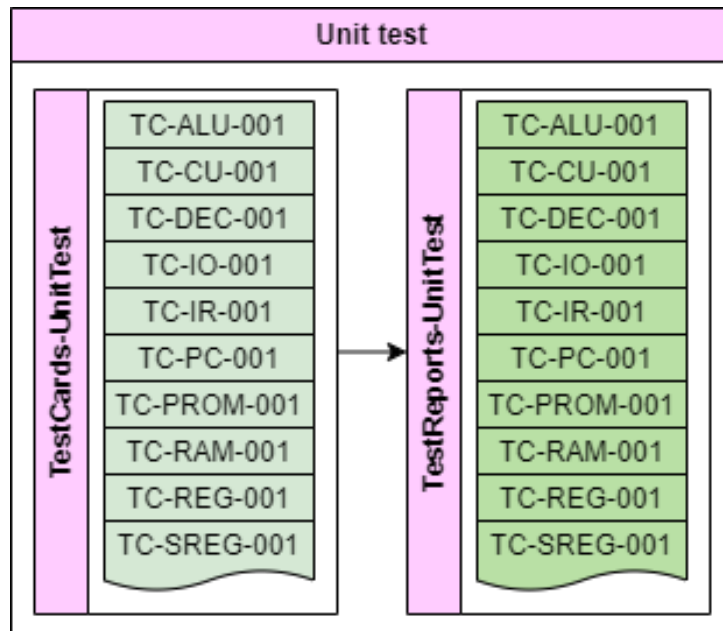


Figure 2: Unit test file structure

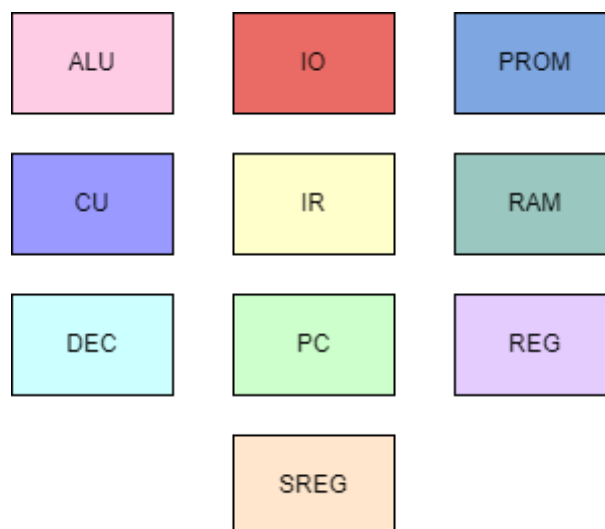


Figure 3: Unit level connection diagram

Figure 2 illustrates the directory organization, where unit testbenches are kept separate from the synthesizable RTL. Figure 3 depicts the generic unit test harness, where a dedicated testbench drives stimulus vectors directly to the Device Under Test (DUT) and monitors outputs, abstracting away the rest of the system.

3.2 Subsystem Tests

Subsystem testing bridges the gap between unit validation and full system integration. In this phase, previously verified units are combined into functional groups to verify interface protocols and data flow. Figure 4 outlines the organization of these intermediate test suites.

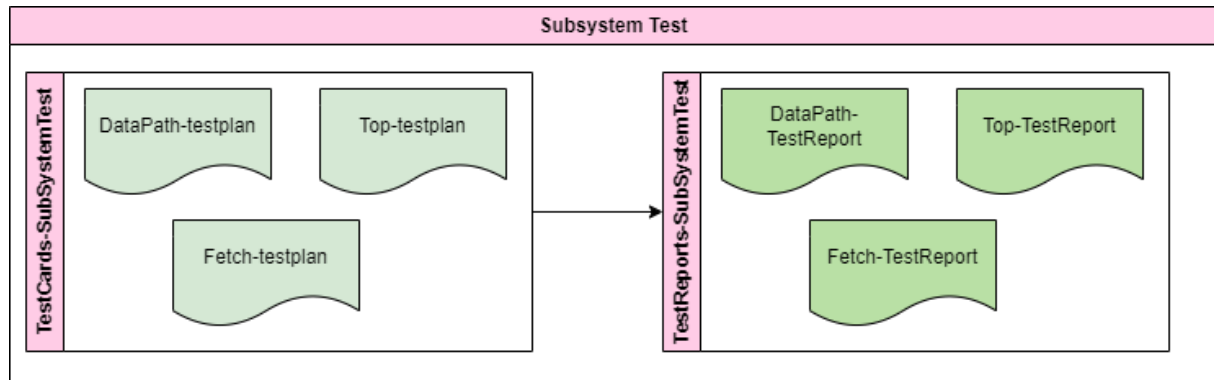


Figure 4: Subsystem test file structure

3.2.1 Stage 2: Data Path Integration (TS-ST-DP-001)

The Data Path integration focuses on the execution core of the processor. As shown in Figure 5, this configuration connects the ALU, Register File, and Status Register (SREG), while bypassing the instruction fetch logic. A test sequencer acts as a surrogate Control Unit, manually driving control signals to verify arithmetic pipelines, flag propagation, and register write-back mechanisms in isolation.

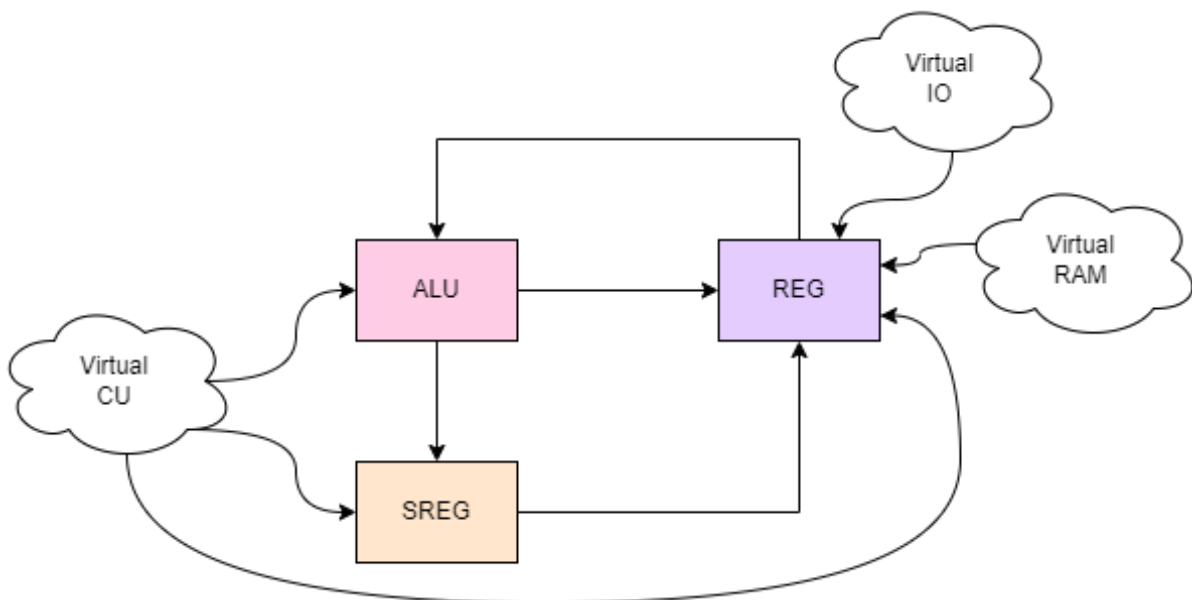


Figure 5: Data Path Integration Test Configuration

3.2.2 Key Integration Points

Interface	Test Focus	Verification Method
Decoder → RegFile	io_addr_in[7:0] to I/O address	I/O register addressing
ALU → RegFile	result[7:0] to write-back	Arithmetic operation sequences
RegFile → ALU	data_a/b[7:0] to operands	Register bypass, operand selection
ALU → SREG	flags[4:0] to status	Flag generation, persistence
SREG → RegFile	sreg_data[7:0] to load	Status restore (POP SREG)
Control → ALU/RegFile	Control signal timing	Synchronization, pipeline hazards

3.3 Stage 3: Fetch Unit Integration (TS-ST-FETCH-001)

The Fetch Unit integration isolates the instruction supply chain. Figure 6 illustrates the loop formed by the Program Counter (PC), Program ROM, and Instruction Decoder. This stage verifies the processor's ability to navigate memory, handle jumps and branches correctly, and decode opcodes into valid control signals, independent of the execution units.

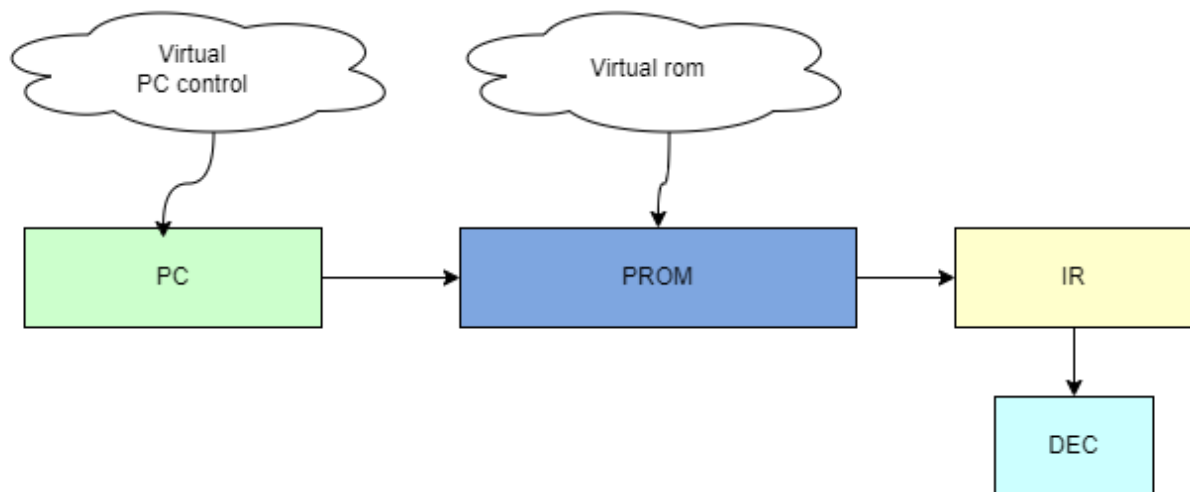


Figure 6: Fetch Unit Integration Test Configuration

3.3.1 Fetch Pipeline Verification

Pipeline Stage	Tested Behavior	Critical Timing
PC Generation	Address sequencing, jumps, branches	Clock cycle 0-2ns
ROM Access	Instruction fetch, timing, wait states	Cycle 2-5ns
IR Latching	Instruction stability, enable timing	Rising edge +1ns
Decoding	Opcode extraction, field parsing	Combinational delay <2ns
Control Handoff	Decoded signals to execution unit	End of cycle validation

3.4 Stage 4: Top-Level System Integration (TS-ST-TOP-001)

Top-level integration assembles the complete PolyFlop8 processor. In this configuration (Figure 7), the Control Unit (FSM) takes over signal generation, orchestrating the interaction between the Fetch and Data Path subsystems. Verification involves running actual assembly code sequences to validate the full instruction cycle—Fetch, Decode, Execute, and Write-Back—and ensuring correct architectural state changes in memory and I/O.

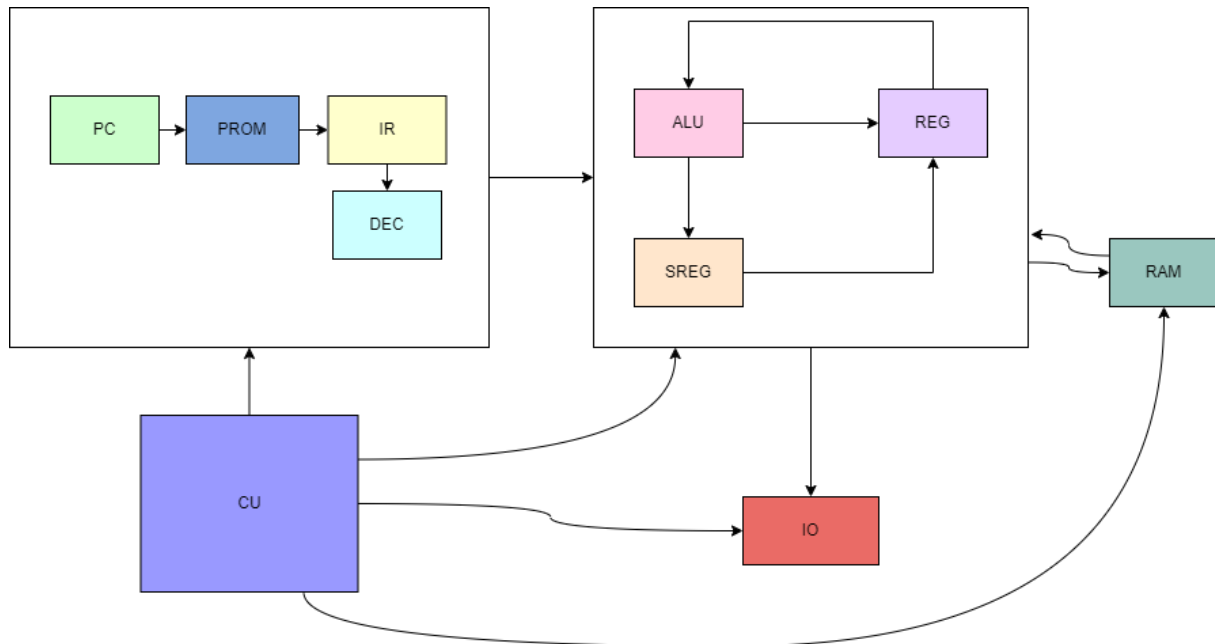


Figure 7: Top-Level System Test Configuration

3.4.1 System-Level Verification Matrix

Test Category	Verification Focus	Instructions Covered	Critical Paths
I/O Operations	CU → I/O → RegFile → External	IN, OUT, SBI, CBI	io_addr_in timing
Arithmetic Pipeline	ALU → RegFile → SREG → Control	ADD, SUB, ADC, SBC	Result propagation
Memory Subsystem	CU → RAM → RegFile → ALU	LD, ST, LDS, STS	Address calculation
Control Flow	CU → PC → ROM → Decoder → CU	RJMP, RCALL, RET, BRxx	Branch prediction
I/O Operations	CU → I/O → RegFile → External	IN, OUT, SBI, CBI	Timing constraints
Exception Handling	Reset, illegal opcodes, time-outs	All instructions	Recovery sequences

4 TEST ENVIRONMENT ARCHITECTURE

4.1 Verification IP (VIP) Components

VIP Component	Purpose	Reuse Level
Clock/Reset Generator	Generate system clock and reset sequences	All levels
Register Access BFM	Read/write register file through all interfaces	Unit, Data Path
Memory Model	Simulate ROM/RAM behavior with backdoor access	All levels
Instruction Sequencer	Generate instruction sequences for testing	Fetch, System
Flag Checker	Monitor and verify status flag behavior	Data Path, System
Pipeline Monitor	Track instruction flow through pipeline stages	System only
Coverage Collector	Collect functional and code coverage metrics	All levels

5 DETAILED TEST PLAN ALLOCATION

5.1 Data Path Test Allocation (TS-ST-DP-001)

Tests allocated to verifying the arithmetic logic pipeline and register integrity.

ID	Test Case Title	Priority
DP-01	Register Read/Write Integrity (RegFile isolation)	P1
DP-02	ALU-Register Loopback (Result propagation)	P1
DP-03	Flag Persistence (SREG retention checks)	P2
DP-04	Carry Propagation (ADC/SBC chain verification)	P2

5.2 Fetch Unit Test Allocation (TS-ST-FETCH-001)

Tests allocated to verifying Program Counter manipulation and Instruction decoding.

ID	Test Case Title	Priority
FT-01	Sequential Fetching (PC increment)	P1
FT-02	Absolute Jump (JMP instruction verification)	P1
FT-03	Relative Jump (RJMP with negative offsets)	P2
FT-04	Branch NOT Taken (Fall-through logic)	P2
FT-05	Reset Behavior (Vector 0x000 enforcement)	P3

5.3 Top Level Test Allocation (TS-ST-TOP-001)

Full system verification executing assembly sequences.

ID	Test Case Title	Priority
SYS-01	"Hello World" (LDI + ADD sequence)	P1
SYS-02	Memory Store & Load (RAM access verification)	P1
SYS-03	Conditional Loop (BRNE counter implementation)	P2
SYS-04	IO Output Test (Peripheral communication)	P2

APPENDIX A: SIGNAL DICTIONARY

Signal	Width	Description	Test Stage
alu_sel	4-bit	ALU operation selection	Unit, DP
rf_src_sel	3-bit	Register file write source	Unit, DP, System
pc_src	2-bit	Program counter source selection	Unit, Fetch, System
mem_addr_sel	2-bit	RAM address source selection	Unit, System
mem_we	1-bit	RAM write enable	Unit, System
regfile_we	1-bit	Register file write enable	All stages
sreg_we	1-bit	Status register write enable	Unit, DP, System
io_we	1-bit	I/O write enable	Unit, System
io_addr_in	8-bit	I/O address input from Decoder to Register File	Unit, System
data_r7x	8-bit	Hardwired R7 output for X-pointer	Unit, DP, System
flags	5-bit	ALU flags to status register	Unit, DP